

BRACT's

Vishwakarma Institute of Technology, Pune

Practical Implementation Sheet

Department: CSE-AI Semester: 5 Academic Year: 2026-27

Division: E Batch: 1 Practical No: 1

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Subject Name: Deep Learning

Problem Statement:

Assignment 1 - Install and configure TensorFlow/Keras in Google Colab. Perform data preprocessing, normalization, train-test split, and visualization on a sample dataset.

Algorithm:

TensorFlow/Keras

1. **Import** the required libraries:
 - TensorFlow/Keras
 - NumPy
 - Matplotlib
2. **Load** the MNIST dataset using `tf.keras.datasets.mnist.load_data()`.
3. **Display** the shape of the training and testing datasets.
4. **Check** the data type and pixel value range of the images.
5. **Normalize** the image pixel values by dividing each pixel by 255.0.
6. **Verify** the normalized pixel value range (0–1).

7. **Use** the predefined training and testing datasets (train-test split already provided by MNIST).
8. **Visualize** a few sample images with their corresponding labels using Matplotlib.

Code:

```
import tensorflow as tf
import matplotlib.pyplot as plt
import numpy as np
(x_train,y_train),(x_test,y_test)=tf.keras.datasets.mnist.load_data()
print("X_Train :",x_train.size)
print("X_Test",y_train.size)
print(x_test.size)
print(y_test.size)
x_train = x_train / 255.0
x_test = x_test / 255.0

print("After Normalization")
print("Minimum:", x_train.min())
print("Maximum:", x_train.max())
print("Training Samples:", len(x_train))
print("Testing Samples:", len(x_test))
plt.figure(figsize=(10,5))

for i in range(10):
    plt.subplot(2,5,i+1)
    plt.imshow(x_train[i], cmap='gray')
    plt.title(f"Label: {y_train[i]}")
    plt.axis('off')

plt.show()
```

Output:

```
X_Train : 47040000  
X_Test 60000  
7840000  
10000  
After Normalization  
Minimum: 0.0  
Maximum: 1.0  
Training Samples: 60000  
Testing Samples: 10000
```

